

Cinema E-Booking System

Term Project Specification

CSCI 4050/6050 – Software Engineering
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Problem Statement

The Cinema E-Booking System is a web-based application designed for a single theater with multiple halls and a limited number of employees. The system manages information about movies, showtimes, ticket pricing, and users.

The system allows users to browse and search movie information, select movies, showtimes, ticket quantities, age categories, and seats prior to checkout. Checkout and payment functionality are available only to authenticated users.

The system maintains user account information and supports registration, authentication, password recovery, and profile management. Personalized features include maintaining a list of favorite movies and receiving movie recommendations.

System administrators manage movie listings, showtimes, ticket prices, promotions, and user accounts, and can view real-time statistics related to system usage and sales.

Ticket prices are determined based on the age category of the ticket holder. The system currently supports Child, Adult, and Senior categories.

High-Level System Requirements

A. Movie and Theater Management (Administrator)

1. The system must allow administrators to add, update, and delete movie information.
2. For each movie, the system must record relevant information including the movie title, category, cast, director, producer, synopsis, reviews, trailer image and video, MPAA-US film rating code, and show dates and times.
3. The system must allow administrators to add, update, and manage ticket prices, online booking fees, and promotional offers.
4. The system must allow administrators to manage user accounts, including adding new administrators, updating or deleting user accounts, and suspending user accounts when necessary.
5. The system must allow administrators to send promotional communications to users who have opted to receive promotions.

6. The system must allow administrators to view real-time statistics related to bookings, sales, currently running movies, and registered users.

B. User Registration and Account Management

7. The system must allow users to create an account by providing required personal information, including name, email address, phone number, and password.
8. Users may optionally provide home address information and payment information. Each user may store one address and up to three payment cards.
9. User registration must be verified using a system-generated verification mechanism delivered to the user through a registered communication channel (e.g., email).
10. The system must provide a secure forgot-password facility.
11. Authenticated users must be able to view and modify their user profile, including personal information, password, preferences, and promotion subscription settings. Rules governing changes to sensitive information may be defined in later development iterations.

C. Browsing, Search, and Ticket Selection

12. The system must allow users to browse movie information.
13. The home page must display currently playing movies and coming-soon movies.
14. The system must provide search and filtering functionality that allows users to search movies by title, category, or show date.
15. Users must be able to select a movie showing, specify the number of tickets and the age category for each ticket, and select corresponding seats using a graphical representation of the theater hall.

D. Checkout, Payment, and Orders

16. Checkout and payment functionality must be available only to authenticated users.
17. The system must provide a secure checkout and payment facility that supports payment using stored user payment information.
18. The system must allow users to apply valid promotion codes during checkout.
19. The system must store order information, including booking number, ticket number, seat identifier(s) associated with each ticket, movie title, show date and time, and payment reference information.
20. The total order price must be calculated based on ticket prices, applicable taxes, online booking fees, and any applied promotional discounts.
21. Authenticated users must be able to view their order history.
22. Users may cancel tickets up to 60 minutes before showtime and receive a refund (low priority).

E. Personalization Features

23. The system must allow authenticated users to add movies to a personal Favorites list, regardless of whether tickets have been purchased for those movies.
24. Authenticated users must be able to view and manage their Favorites list, including adding or removing movies.

- 25. The system must provide a Recommendation Service that suggests movies to users.
- 26. Recommendations may be based on user-selected favorite movies, summarized user preferences, or aggregated system activity.
- 27. The Recommendation Service must operate as a separate system component with a clearly defined interface.
- 28. When integrating external services, teams must not transmit personal identifiers, payment data, or raw transaction records. Only summarized, non-identifiable information may be shared.

F. System Design and Technical Requirements

- 29. The system must support multi-user access and ensure correct concurrent behavior.
- 30. The system must enforce authentication and authorization for different user roles.
- 31. The system must apply appropriate security controls, including encryption for sensitive data (user passwords and payment card numbers), in accordance with accepted web application security practices.
- 32. The system must provide an easy-to-use user interface that conforms to authorization rules.
- 33. The system must be accessible through common web browsers.
- 34. The system must use a persistent data store.
- 35. The system must be implemented using an object-oriented programming approach.
- 36. The system design must follow a separation-of-concerns architecture. Teams are expected to structure their systems using a multi-layered design with clearly defined interfaces between layers, aligned with MVC principles, or an equivalent hybrid multi-layer architecture.
- 37. The system should use accepted software and web standards whenever possible.